



NASIR MAHMOOD ABBASI, PhD

Bioinformatician

+33 7 66 03 75 20 (mobile within France) | nmabbasi@gmail.com

nmabbasi.github.io | linkedin.com/in/nmabbasi | 0009-0005-0083-1984

PROFILE

Bioinformatician with 9+ years across genomics, transcriptomics, and single-cell analysis, including a PhD applying single-cell multi-omics to cancer heterogeneity. Comfortable owning an analysis end-to-end or integrating into a larger team.

- Builds reproducible analysis pipelines, from raw sequencing data through statistical validation
- Communicates technical findings clearly to non-computational and clinical collaborators
- Autonomous on open-ended problems; strong project management and prioritization skills

PROFESSIONAL EXPERIENCE

PhD Researcher, Université de Bordeaux / INSERM U1312 (BRIC)

Sep 2023 – Jun 2026

Bordeaux, France

- Built a single-cell multi-omic pipeline to characterize tumor heterogeneity in a rare lymphoma
- Combined WES, optical genome mapping, and single-cell transcriptomics into one coherent model
- Worked with wet-lab colleagues to validate a computational prediction experimentally

Research Engineer, IRBI Insect Biology Research Institute

Jun 2022 – Aug 2023

Tours, France

- Administered the group's HPC/Slurm cluster and long-read sequencing workflows

Research Engineer, INRAE

Mar 2019 – Apr 2021

Clermont-Ferrand, France

- Benchmarked small RNA-seq tools and set the group's standard analysis pipeline

Research Assistant, Hong Kong University of Science and Technology (HKUST)

Jan 2017 – Apr 2018

Hong Kong, China

- Benchmarked exome/whole-genome pipelines for variant calling in cancer patient data

Earlier Research Engineer and Research Assistant roles

2009 – 2017

Universidad Miguel Hernández de Elche | Evolutionary Biology Centre | Karolinska Hospital | COMSATS University

- Marine prokaryotic genomics, behavioral genetics, bacterial genomics, and rare disease exome sequencing

EDUCATION

PhD in Cancer Biology, Université de Bordeaux

2023 – 2026

INSERM U1312, Bordeaux Research in Translational Oncology (BRIC)

MSc in Bioinformatics, Uppsala University

2010 – 2012

Sweden

LANGUAGES

English: Fluent | French: Professional working proficiency | Urdu: Native

TECHNICAL SKILLS

Single-cell analysis: Seurat, Scanpy, SingleCellExperiment, SCTransform, Harmony, DoubletFinder, Scrublet, Azimuth, SingleR, scType, Monocle3, Slingshot, PAGA, scVelo, Velocyto

Spatial transcriptomics: Seurat spatial, squidpy (self-directed, in progress)

Genomics: WES, WGS, RNAseq, Metatranscriptomics, Ampicon sequencing, BioNano optical mapping, variant calling, inferCNV, SCEVAN

Regulatory & communication networks: SCENIC, GENIE3, DoRothEA, LIANA, CellChat

Statistics & differential expression: DESeq2, edgeR, MAST, LIBRA (pseudobulk), Wilcoxon rank-sum test

Visualization: ggplot2, ComplexHeatmap, SCpubr, Nebulosa, dittoSeq

Languages: R, Python, Perl, C, JavaScript, PHP, SQL

Infrastructure & tools: HPC/Slurm administration, Snakemake, Conda, Docker, Singularity, Git, Galaxy administration, cloud computing

Under Review

Boccelari L, Gravelle P, Tosolini M, **Abbasi N**, Le Morvan V, et al. Primary Cutaneous Diffuse Large B Cell Lymphoma, Leg Type, Induces Dynamic Transcriptional Reprogramming of Immunosuppressive Macrophages. *Cancer Immunology Research*. Under review

In Preparation

Abbasi N, et al. Deciphering Sézary syndrome tumoral heterogeneity through a bioinformatic approach. *To be submitted online, end of 2026*

Published

Ajmal M, Khan M, Neveling K, Tayyab A, Jaffar S, Sadeque A, Ayub H, **Abbasi N**, Riaz M, Micheal S, Gilissen C, Ali S, Azam M, Collin R, Cremers F, Qamar R. Exome sequencing identifies a novel and a recurrent BBS1 mutation in Pakistani families with Bardet-Biedl syndrome. *Molecular Vision*. 2013;19:644-653

Zangenah S, **Abbasi N**, Anders FA, Bergman P. Whole Genome Sequencing Reveals Novel Species of genus Capnocytophaga Isolated From Dog and Cat Bite Wounds in Humans. *Scientific Reports*. 2016

Zajitschek S, Herbert-Read JE, Abbasi N, Immler S. Paternal personality and social status influence offspring personality in offsprings. *BMC Evolutionary Biology*. 2017;17:157